

6 The Inductive Nature of Arithmetic

Recall that we introduced a new, compact notation for expressing a common class of first-order sentences. We will now generalize this notation to any finite number of variables that are *all quantified the same way*.

Definition 6.1: Restricted Quantifier Notation.

Given a set X and a positive integer n , we make the following definitions.¹

$$\begin{aligned} (\forall x_1, \dots, x_n \in X)(\varphi(x)) &:= \forall x_1 \dots \forall x_n ((x_1 \in X \wedge \dots \wedge x_n \in X) \Rightarrow \varphi(x)) \\ (\exists x_1, \dots, x_n \in X)(\varphi(x)) &:= \exists x_1 \dots \exists x_n ((x_1 \in X \wedge \dots \wedge x_n \in X) \wedge \varphi(x)) \end{aligned}$$

¹We are using the phrase “positive integer” here in an *intuitive* way to denote the number of variables being quantified. The numbers appearing in subscripts—such as x_1 and x_2 —are just ways of distinguishing variables from each other.

6.1 The Least Element Principle

Definition 6.2: Order on $\mathbb{N}$.

Given $x, y \in \mathbb{N}$, we define what it means for x to be *(strictly) less than* y as follows.

$$x < y := x \in y$$

Similarly, for any $x, y \in \mathbb{N}$, we define what it means for x to be *less than or equal to* y .²

$$x \leq y := (x < y) \vee (x = y)$$

We further define $x > y := y < x$ and $x \geq y := y \leq x$ for every $x \in \mathbb{N}$ and $y \in \mathbb{N}$ as you might expect. These definitions establish a *strict total order* $<$ on $\mathbb{N}$, along with a corresponding *weak total order* $\leq$, as we will see in the theorem below.

Theorem 6.1: The Natural Numbers are Linearly Ordered by $\leq$.

The order $\leq$ defined on the set $\mathbb{N}$ has the following four properties.

- | | |
|--|--------------|
| 1. $(\forall x \in \mathbb{N})(x \leq x)$. | Reflexivity |
| 2. $(\forall x, y \in \mathbb{N})((x \leq y \wedge y \leq x) \Rightarrow x = y)$. | Antisymmetry |
| 3. $(\forall x, y, z \in \mathbb{N})((x \leq y \wedge y \leq z) \Rightarrow x \leq z)$. | Transitivity |
| 4. $(\forall x, y \in \mathbb{N})(x \leq y \vee y \leq x)$. | Totality |

The first three properties make $\mathbb{N}$ with $\leq$ a *partially ordered* set;³ the inclusion of the last property makes $\mathbb{N}$ with $\leq$ a *linearly ordered* set.

In fact, the order $\leq$ we just defined has one more remarkable property: whenever we look at a nonempty set of natural numbers—call it M , where $M \subseteq \mathbb{N}$ and $M \neq \emptyset$ —we can always find an element $m \in M$ that is *smallest* according to the order $\leq$ we defined. To say that m is the *smallest element* of M formally, we would say that m is “smaller than anything else in M ,” which we could express as $(\forall n \in M)(m \leq n)$.

Theorem 6.2: Least Element Property of $\mathbb{N}$.

$$\forall M((M \subseteq \mathbb{N} \wedge M \neq \emptyset) \Rightarrow (\exists m \in M)(\forall n \in M)(m \leq n)).$$

This theorem is written with our new notation. To be clear, this theorem is equivalent to the following sentence: $\forall x((x \subseteq \mathbb{N} \wedge x \neq \emptyset) \Rightarrow \exists y(y \in x \wedge \forall z(z \in x \Rightarrow y \leq z)))$. This immensely important theorem tells us that $\mathbb{N}$ is *well-ordered* by $\leq$, and that makes it possible for us to prove properties of natural numbers in an *interesting* way.

²Whenever we define a *strict* order $<$ on a set X , we can always define a *weak* version $\leq$ of that order by saying $x \leq y := x < y \vee x = y$ for all $x, y \in X$.

³We say a relation $\leq$ on a set X is a *partial order* by definition *iff* the relation is *reflexive*, *antisymmetric*, and *transitive*; i.e., the relation $\leq$ satisfies the first three properties listed in *Theorem 6.1*.

Theorem 6.3: Weak Induction.

For any wff φ with at most one free variable, the following equivalence holds.

$$(\forall n \in \mathbb{N})(\varphi(n)) \Leftrightarrow \varphi(0) \wedge (\forall k \in \mathbb{N})(\varphi(k) \Rightarrow \varphi(\text{succ}(k)))$$

Proof. Let φ be a wff. We will prove the equivalence by showing both directions.

Fragment 1: Forwards.

Assume $(\forall n \in \mathbb{N})(\varphi(n))$. This immediately implies $\varphi(0)$ since $0 \in \mathbb{N}$. For the rest of the proof, let $k \in \mathbb{N}$ and assume $\varphi(k)$. Observe that $\varphi(\text{succ}(k))$ because we assumed $(\forall n \in \mathbb{N})(\varphi(n))$. Therefore, $\varphi(k) \Rightarrow \varphi(\text{succ}(k))$, and we can thus conclude $(\forall n \in \mathbb{N})(\varphi(n) \Rightarrow \varphi(\text{succ}(n)))$ as desired.

Fragment 2: Backwards.

Assume $\varphi(0)$ and $(\forall k \in \mathbb{N})(\varphi(k) \Rightarrow \varphi(\text{succ}(k)))$. Towards a contradiction, suppose there exists $n \in \mathbb{N}$ such that $\neg\varphi(n)$. Consider $M := \{x \mid x \in \mathbb{N} \wedge \neg\varphi(x)\}$, the set of all counterexamples to φ . Because we assumed $\neg\varphi(n)$ and $n \in \mathbb{N}$, we know that $n \in M$ by definition, so $M \neq \emptyset$. Further, $M \subseteq \mathbb{N}$ because every element of M must be a natural number. Thus, we know that there exists some $m \in M$ such that $(\forall x \in M)(m \leq x)$ by **Theorem 6.1**. Since $m \in M$, we know $m \in \mathbb{N}$ ⁴ and $\neg\varphi(m)$.

Case 1:

Suppose $m = 0$ and recall that $\neg\varphi(m)$ because $m \in M$. We know $\varphi(0)$ by assumption, which means we know $\varphi(m)$, contradicting $\neg\varphi(m)$. $\nexists$

Case 2:

Suppose that there exists $\ell \in \mathbb{N}$ such that $m = \text{succ}(\ell)$. By definition, this means $m = \text{succ}(\ell) = \ell \cup \{\ell\}$. Because $\ell \in \{\ell\}$, we know $\ell \in \ell \cup \{\ell\}$, so $\ell \in m$. Since we know $\forall x(x \notin x)$, we can say $\ell \neq m$. It is at this point that we must recall that m is the least element of M , which means $(\forall x \in M)(m \leq x)$, implying $m \leq \ell$. This means $m < \ell \vee m = \ell$ by definition. However, we know $\ell \neq m$, so we have $m < \ell$, which means $m \in \ell$ by definition. We now have both $\ell \in m$ and $m \in \ell$, contradicting the theorem $\forall x \forall y(x \in y \Rightarrow y \notin x)$. $\nexists$

We have contradictions in both cases, so we instead conclude $\varphi(n)$ as desired.

Having proven both directions, we are now burdened with the weight of knowing that $(\forall n \in \mathbb{N})(\varphi(n)) \Leftrightarrow \varphi(0) \wedge (\forall k \in \mathbb{N})(\varphi(k) \Rightarrow \varphi(\text{succ}(k)))$.

QED

Lemma 6.1.

$$(\forall x \in \mathbb{N})(\text{succ}(x) \neq 0).$$

Proof. Let $x \in \mathbb{N}$. Notice that $x \in \{x\}$, so that $x \in x \cup \{x\}$, so that $x \in \text{succ}(x)$ by definition. Therefore, $\text{succ}(x) \neq \emptyset$, which means $\text{succ}(x) \neq 0$ by definition.

QED

⁴Recall there are two kinds of natural numbers: zero and the ones that are successors.

6.2 Addition

Definition 6.3: Addition on the Natural Numbers.

We define how to *add* a natural number $n \in \mathbb{N}$ with another one as follows.

$$\begin{aligned} n + 0 &:= n \\ n + \text{succ}(m) &:= \text{succ}(n + m) \quad \text{if } m \in \mathbb{N} \end{aligned}$$

Lemma 6.2.

$$1 + 1 = 2.$$

Proof. Observe the following sequence of equalities.

$$\begin{aligned} 1 + 1 &= 1 + \text{succ}(0) && \text{by definition of 1} \\ &= \text{succ}(1 + 0) && \text{by definition of +} \\ &= \text{succ}(1) && \text{by definition of +} \\ &= 2 && \text{by definition of 2} \end{aligned}$$

Therefore, $1 + 1 = 2$. Rejoice.

QED

Lemma 6.3.

$$(\forall x \in \mathbb{N})(x + 0 = x).$$

Proof. Let $x \in \mathbb{N}$ and observe that $x + 0 = x$ by definition of addition on $\mathbb{N}$.

QED

Lemma 6.4.

$$(\forall x \in \mathbb{N})(0 + x = x).$$

Proof. We will show $(\forall n \in \mathbb{N})(0 + n = n)$ by *weak induction*.

Basis Step:

Observe that $0 + 0 = 0$ by definition of addition on $\mathbb{N}$.

Inductive Step:

Let $k \in \mathbb{N}$, and assume that $0 + k = k$. We will show $0 + \text{succ}(k) = \text{succ}(k)$. Observe.

$$\begin{aligned} 0 + \text{succ}(k) &= \text{succ}(0 + k) && \text{by definition of +} \\ &= \text{succ}(k) && \text{by the inductive hypothesis} \end{aligned}$$

Therefore, we can conclude $(\forall n \in \mathbb{N})(0 + n = n)$ as desired.

QED

Lemma 6.5.

$$(\forall x \in \mathbb{N})(x + 1 = \text{succ}(x)).$$

Proof. Let $x \in \mathbb{N}$. Observe the following.

$$\begin{aligned} x + 1 &= x + \text{succ}(0) && \text{by definition of 1} \\ &= \text{succ}(x + 0) && \text{by definition of +} \\ &= \text{succ}(x) && \text{by definition of +} \end{aligned}$$

Therefore, $x + 1 = \text{succ}(x)$ as desired.

QED

Theorem 6.4: Associativity of Addition.

$$(\forall x, y, z \in \mathbb{N})(x + (y + z) = (x + y) + z).$$

Proof. Let $x, y \in \mathbb{N}$. We prove $(\forall z \in \mathbb{N})(x + (y + z) = (x + y) + z)$ by *weak induction*.

Basis Step:

Notice that $x + (y + 0) = x + y = (x + y) + 0$ by definition of addition on $\mathbb{N}$.

Inductive Step:

Let $k \in \mathbb{N}$ and assume that $x + (y + k) = (x + y) + k$. Observe now the following.

$$\begin{aligned} x + (y + \text{succ}(k)) &= x + \text{succ}(y + k) && \text{by definition of } + \\ &= \text{succ}(x + (y + k)) && \text{by definition of } + \\ &= \text{succ}((x + y) + k) && \text{by the inductive hypothesis} \\ &= (x + y) + \text{succ}(k) && \text{by definition of } + \end{aligned}$$

We thus have $x + (y + \text{succ}(k)) = (x + y) + \text{succ}(k)$.

We can therefore conclude that $(\forall z \in \mathbb{N})(x + (y + z) = (x + y) + z)$ as desired.

QED

Theorem 6.5: Commutativity of Addition.

$$(\forall x, y \in \mathbb{N})(x + y = y + x).$$

Proof. This proof is left as an exercise to the reader. □

6.3 Multiplication

Definition 6.4: Multiplication on the Natural Numbers.

We define how to *multiply* a natural number $n \in \mathbb{N}$ with another one as follows.

$$\begin{aligned} n \cdot 0 &:= 0 \\ n \cdot \text{succ}(m) &:= (n \cdot m) + n \quad \text{if } m \in \mathbb{N} \end{aligned}$$

Lemma 6.6: One is the Multiplicative Identity.

$$(\forall x \in \mathbb{N})((x \cdot 1 = x) \wedge (1 \cdot x = x)).$$

Proof. First, let $n \in \mathbb{N}$ and observe that $n \cdot 1 = n \cdot \text{succ}(0) = (n \cdot 0) + n = 0 + n$ by the definition of multiplication. Recalling that addition is commutative, we can then see $0 + n = n + 0 = n$ by the definition of addition. Thus, $(\forall x \in \mathbb{N})(1 \cdot x = x)$.

Next, we will prove $(\forall x \in \mathbb{N})(x \cdot 1 = x)$ by *weak induction*.

Basis Step:

Observe that $1 \cdot 0 = 0$ by the definition of multiplication.

Inductive Step:

Let $k \in \mathbb{N}$ and assume $1 \cdot k = k$. We will now show $1 \cdot \text{succ}(k) = \text{succ}(k)$. Observe.

$$\begin{aligned} 1 \cdot \text{succ}(k) &= (1 \cdot k) + 1 && \text{by definition of multiplication} \\ &= k + 1 && \text{by the inductive hypothesis} \\ &= \text{succ}(k) && \text{because } (\forall x \in \mathbb{N})(\text{succ}(x) = x + 1) \end{aligned}$$

Therefore, we can conclude that $(\forall x \in \mathbb{N})(x \cdot 1 = x)$.

QED

Lemma 6.7: Zero is the Multiplicative Annihilator.

$$(\forall x \in \mathbb{N})(x \cdot 0 = 0) \wedge (0 \cdot x = 0).$$

Proof. First, let $n \in \mathbb{N}$ and observe that $n \cdot 0 = 0$ by the definition of multiplication. Thus, $(\forall x \in \mathbb{N})(x \cdot 0 = 0)$. Next, we will show $(\forall x \in \mathbb{N})(0 \cdot x = 0)$ by *weak induction*.

Basis Step:

Observe that $0 \cdot 0 = 0$ by the definition of multiplication.

Inductive Step:

Let $k \in \mathbb{N}$ and assume $0 \cdot k = 0$. We will now show $0 \cdot \text{succ}(k) = 0$. Observe.

$$\begin{aligned} 0 \cdot \text{succ}(k) &= (0 \cdot k) + 0 && \text{by definition of multiplication} \\ &= 0 + 0 && \text{by the inductive hypothesis} \\ &= 0 && \text{by definition of addition} \end{aligned}$$

Thus, we have that $0 \cdot \text{succ}(k) = 0$ as desired.

Therefore, we can conclude that $(\forall x \in \mathbb{N})(0 \cdot x = 0)$.

QED

Theorem 6.6: Distributivity of Multiplication over Addition.

$$(\forall x, y, z \in \mathbb{N})(x \cdot (y + z) = (x \cdot y) + (x \cdot z)).$$

Proof. Left as an exercise for the reader. □

Theorem 6.7: Associativity of Multiplication.

$$(\forall x, y, z \in \mathbb{N})(x \cdot (y \cdot z) = (x \cdot y) \cdot z).$$

Proof. Let $x, y \in \mathbb{N}$. We will prove $(\forall z \in \mathbb{N})(x \cdot (y \cdot z) = (x \cdot y) \cdot z)$ by *weak induction*.

Basis Step:

Notice that $x \cdot (y \cdot 0) = x \cdot 0 = 0$ by the definition of multiplication on $\mathbb{N}$. Similarly, we can see that $(x \cdot y) \cdot 0 = 0$ by the same definition. Thus, $x \cdot (y \cdot 0) = (x \cdot y) \cdot 0$.

Inductive Step:

Let $k \in \mathbb{N}$, and assume $x \cdot (y \cdot k) = (x \cdot y) \cdot k$. We now observe with our eyeballs.

$$\begin{aligned} x \cdot (y \cdot \text{succ}(k)) &= x \cdot ((y \cdot k) + y) && \text{by definition of } \cdot \\ &= x \cdot (y \cdot k) + x \cdot y && \text{by distributivity of } \cdot \text{ over } + \\ &= (x \cdot y) \cdot k + x \cdot y && \text{by the inductive hypothesis} \\ &= (x \cdot y) \cdot \text{succ}(k) && \text{by definition of } \cdot \end{aligned}$$

As a result, we have that $x \cdot (y \cdot \text{succ}(k)) = (x \cdot y) \cdot \text{succ}(k)$.

Therefore, we can conclude $(\forall z \in \mathbb{N})(x \cdot (y \cdot z) = (x \cdot y) \cdot z)$ as desired.

QED

Theorem 6.8: Commutativity of Multiplication.

$$(\forall x, y \in \mathbb{N})(x \cdot y = y \cdot x).$$

Proof. This proof is left as an exercise to the reader.

QED

6.4 Higher-Order Operations

Definition 6.5: Exponentiation.

We define how to *exponentiate* any $n \in \mathbb{N}$ with a natural number exponent as follows.

$$\begin{aligned} n^0 &:= 1 \\ n^{\text{suc}(m)} &:= n^m \cdot n \quad \text{if } m \in \mathbb{N} \end{aligned}$$

Theorem 6.9.

$$(\forall x, y, z \in \mathbb{N})(x^y \cdot x^z = x^{y+z}).$$

Definition 6.6: Tetration.

We define how to raise any $n \in \mathbb{N}$ to a natural number exponent recursively below.

$$\begin{aligned} n \uparrow\uparrow 0 &:= 1 \\ n \uparrow\uparrow \text{suc}(m) &:= n^{n \uparrow\uparrow m} \quad \text{if } m \in \mathbb{N} \end{aligned}$$